Bisection Method: Short Report

Findings on the bisection method from the Project 4 nonlinear equation solver experiments.

Summary

Bisection is slower than Newton's method but far safer. Given an interval [a, b] where f(a) and f(b) have opposite signs and f is continuous, it always closes in on a root, and the half-width of the interval is a guaranteed error bound. The error bound halves at every step (linear convergence). The solver rejects invalid input with a ValueError. The main weaknesses are the number of iterations and a stopping rule based on |f(c)|, which can stop early on very flat functions.

1. Speed and Convergence

On $f(x) = x^2 - 2$ (exact root $\sqrt{2} = 1.41421356237310$) with the interval [1, 2] and tolerance 10^{-10}, bisection converged in 29 iterations to 1.41421356238425, a final error of 1.12×10^{-11}. The error bound $(b - a)/2$ is exactly halved at every step: 0.5, 0.25, 0.125, … down to 1.86×10^{-9} at iteration 29. This is linear convergence, so each iteration gains about one binary digit. For comparison, Newton's method needed only 4 iterations on the same function.

The actual error does not fall smoothly. It went 0.086, 0.164, 0.039, 0.023, 0.008 over the first five midpoints, because a midpoint can land close to or far from the root by chance. Only the interval half-width is guaranteed to decrease.

2. Effect of the Interval and Tolerance

Bisection has no starting-point problem, only a bracket. A wider bracket costs about $\log_2$ of the width ratio in extra iterations (tolerance 10^{-10}, $f(x) = x^2 - 2$):

Interval	Width	Iterations	Final error
[1, 2]	1	29	1.12×10^{-11}
[0, 3]	3	29	1.12×10^{-11}
[0, 10]	10	30	1.12×10^{-11}
[0, 100]	100	31	1.12×10^{-11}
[0, 1000]	1000	32	1.12×10^{-11}

Making the bracket 10 times wider adds only 1 iteration here, so a rough bracket is cheap. The tolerance matters more. On [1, 2], tolerances of 10^{-4}, 10^{-6}, 10^{-8}, 10^{-10} and 10^{-12} took 13, 21, 27, 29 and 38 iterations. The jumps are uneven because the solver stops on $|f(c)| < \text{tol}$, not on the interval width.

3. Failure Cases

- **No sign change:** $x^2 + 1$ on $[-1, 1]$ has $f(a) \cdot f(b) > 0$, so the solver raises a ValueError before iterating. The same happens for $x^3 - 2x + 2$ on $[0, 1]$ and for even-multiplicity roots such as x^2 on $[-1, 2]$, because the sign does not change.
- **Bad interval order:** solving $f(x) = x$ on $(2, 1)$ raises a ValueError because the solver requires $a < b$. Non-finite endpoints or values are rejected the same way.
- **Root at an endpoint:** $x - 1$ on $[1, 3]$ returns immediately with 0 iterations and the message "f(a) is exactly zero".
- **Iteration limit:** $x^2 - 2$ on $[1, 2]$ with max_iter = 10 returns converged = False after 10 iterations, with the root bracketed to within 9.77×10^{-4}.

Function	Interval	Converged	Iterations	Outcome
$x^2 - 2$	[1, 2]	Yes	29	Root found, error 1.1×10^{-11}
$x^3 - 2x + 2$	$[-2, -1]$	Yes	33	Root -1.76929235423
arctan(x)	$[-10, 100]$	Yes	39	Root found (Newton diverged for large x_0)
$x^2 - 1$	$[-3, 0.5]$	Yes	34	Root -1 ($x_0 = 0$ is no issue)
tan(x)	[1, 2]	No	100	Converged to a pole, not a root
x^9	$[-1, 2]$	"Yes"	3	Stopped at 0.125; true root is 0

Pole: tan(x) changes sign on [1, 2], but only because it jumps through a pole at π/2. The solver converges toward 1.5708 with |f| near 10^{16} and reports failure after 100 iterations. A sign change guarantees a root only if f is continuous.

Early stop on a flat function: for x^9 on $[-1, 2]$ with tolerance 10^{-8}, the solver stopped at $x = 0.125$ because $|f(0.125)| = 7.5 \times 10^{-9}$ is below the tolerance, although the true root is 0. The stopping rule $|f(c)| < tol$ does not guarantee a small error in x. Checking the half-width $(b - a)/2$ as well would fix this.

Cases where bisection beats Newton: arctan from far away, $x^3 - 2x + 2$ (where Newton cycled between 0 and 1), and $x^2 - 1$ (where Newton hit a zero derivative) all converge, since bisection needs no derivative.

Conclusion

Bisection is slow but dependable: about 30 iterations for 10^{-10} accuracy, against 4 for Newton, but it never diverges or cycles while the bracket is valid and f is continuous. It needs a sign-changing bracket, it cannot find roots where the sign does not change, and its $|f(c)|$ stopping rule can be too loose on flat functions. A good strategy is to run a few bisection steps to get close, then switch to Newton for fast final convergence.